

Intrinsic Token Weighting for Credit Assignment in Language Model Reinforcement Learning

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Abstract

Policy gradient methods for LLM reinforcement learning rely on value function critics, but language generation prefixes are non-Markov and poorly suited for value estimation, often introducing bias. Recent critic-free approaches such as GRPO and RLOO eliminate value networks but still assign trajectory rewards uniformly across tokens, leaving the credit assignment problem unresolved. In this work, we study token-level credit redistribution without value functions, using intrinsic token-weighting schemes derived from the model’s predictive distribution, including surprisal- and entropy-based weights. These methods require no additional networks, no tree construction, and no reward models. Through experiments on verifiable math and code tasks, we show that intrinsic weighting reduces gradient variance, stabilizes training, and yields interpretable credit assignment while matching established critic-free baselines. Our results demonstrate that redistributing credit via intrinsic distributional signals is a viable alternative to advantage estimation in language RL.

1 Introduction

Reinforcement learning has become an important component of large language model (LLM) training pipelines, particularly for alignment, preference optimization, and reasoning tasks. Most contemporary approaches rely on policy gradient methods augmented with advantage estimation, typically implemented via learned value functions or generalized advantage estimation. These techniques originate from classical reinforcement learning settings in which states are well-defined, approximately Markov, and admit stable value estimates.

Applying this framework to language generation introduces a fundamental mismatch. In autoregressive LLMs, the ‘state’ is a partial sequence of tokens. Such prefixes are not Markov, do not summarize progress toward task completion, and can vary widely in semantic meaning despite superficial similarity. Consequently, estimating a value function over prefixes is both statistically challenging and conceptually ill-defined. Empirically, value heads in language RL are often brittle, sensitive to distributional shift, and prone to introducing bias that outweighs their variance-reduction benefits.

Motivated by these issues, recent work has increasingly moved away from learned critics in LLM reinforcement learning. Critic-free, outcome-based methods such as RLOO and GRPO replace value estimation with batch-level baselines, achieving stable training on verifiable reasoning tasks such as math and code. However, these approaches still implicitly assume uniform credit assignment across tokens: every action in a trajectory receives the same reward signal, modulated only by a scalar baseline.

This raises a natural question: *if we no longer estimate values over prefixes, how should trajectory-level reward be distributed across tokens?* Uniform credit assignment is simple, but it ignores the fact that different tokens play very different roles in generation. Some tokens correspond to high-uncertainty decisions, strategy commitments, or branching points in reasoning, while others are low-entropy continuations or stylistic filler. Treating all tokens equally risks diluting the learning signal and obscuring the mechanisms by which policies improve.

In this work, we study token-level credit redistribution as a mechanism for on-policy LLM reinforcement

learning without value functions. Rather than estimating advantages or prefix values, we propose to weight token-level policy gradients using intrinsic signals derived directly from the model’s predictive distribution. These include token surprisal and changes in predictive entropy, which naturally reflect decision difficulty and commitment during generation. Importantly, these signals are inexpensive to compute, require no additional networks or tree structures, and make no assumptions about the existence of a meaningful value function over prefixes.

We conduct a systematic empirical evaluation of intrinsic token-weighting schemes on verifiable math and code reasoning tasks. We compare uniform outcome-based policy gradients, batch-baseline methods such as RLOO/GRPO, and several intrinsic weighting strategies under otherwise identical training conditions. Our results show that intrinsic token weighting can reduce gradient variance, stabilize optimization, and produce interpretable credit assignment patterns, while achieving performance comparable to established critic-free baselines.

Overall, this paper reframes advantage estimation as one instantiation of a broader credit assignment problem in language reinforcement learning. We demonstrate that, in settings where state semantics are weak and rewards are trajectory-level, redistributing credit using intrinsic distributional signals is a practical and effective alternative to value-based methods.

2 Related Work

2.1 From RLHF to RLVR

Modern language-model post-training inherits its optimization machinery from policy-gradient reinforcement learning, from REINFORCE [25] to trust-region and clipped-surrogate methods such as TRPO and PPO [16, 18]. Advantage estimation, especially GAE, became the standard variance-reduction device in continuous-control RL by learning value functions over states or prefixes [17]. In language modeling, these ideas entered mainstream use through RLHF systems that optimize sequence-level rewards derived from human preferences or reward models [5, 21, 14].

The recent reasoning-model wave shifted part of the field from preference-based RLHF toward reinforcement learning with verifiable rewards (RLVR), where supervision is often a deterministic correctness signal on math or code tasks. DeepSeekMath introduced GRPO as a practical critic-free alternative for these settings [19], and DeepSeek-R1 popularized large-scale RL-only or RL-dominant reasoning pipelines [7]. Subsequent surveys document how quickly this regime became the default template for open reasoning-model replication and extension [29]. This transition matters for our setting because RLVR makes sparse outcome rewards and long reasoning trajectories central, thereby exposing the credit-assignment problem more directly than earlier short-form RLHF tasks.

2.2 Critic-Based and Critic-Free Policy Optimization

The classical solution to delayed reward is to learn a critic or value function and convert returns into token- or prefix-level advantages. PPO-style RLHF pipelines follow this recipe, but the value-estimation problem is particularly awkward for text generation because prefixes are high-dimensional, non-Markov, and semantically heterogeneous. A growing body of work therefore revisits whether value heads are necessary at all in LLM post-training.

ReMax replaces learned critics with a greedy baseline and shows that much of PPO’s practical benefit can be retained with a simpler REINFORCE-style objective [12]. Back to Basics systematically compares PPO with RLOO-style estimators and finds that critic-free methods can match or exceed value-based baselines in RLHF [1]. REINFORCE++ further strengthens this line by improving robustness to prompt and reward-model variation without reintroducing a learned value function [9]. In reasoning-focused RLVR, GRPO normalizes rewards across a group of sampled responses rather than learning prefix values [19], and later work refines this family with theoretical analyses, off-policy variants, and sequence-level formulations such as GSPO [13, 30]. These methods substantially simplify the optimization stack, but they generally still broadcast a trajectory-level signal across all tokens once a scalar baseline is fixed.

2.3 Reasoning-Specific Analyses of RLVR

Another recent line asks why critic-free RLVR works so well for reasoning in the first place. A Minimalist Approach to LLM Reasoning argues that strong reasoning gains can emerge from a comparatively simple recipe that moves from rejection sampling to REINFORCE-style online updates, suggesting that some of the field’s complexity is optional rather than essential [26]. Complementarily, Wen et al. analyze RLVR theoretically and empirically, arguing that answer-level verifiable rewards can nonetheless incentivize correct intermediate reasoning early in a trajectory [24]. These results are important because they show that uniform outcome-level objectives already contain nontrivial reasoning signal.

At the same time, they do not eliminate the question of *which* tokens should absorb that signal. Recent empirical analysis of RLVR training dynamics suggests that improvements are disproportionately driven by a minority of high-entropy “forking” tokens, and that restricting updates to this subset can remain competitive or even outperform full-token updates in some settings [23]. This is the closest recent line to our work conceptually: it supports the claim that token salience is highly non-uniform during reasoning RL. Our work differs in scope and mechanism, however, by studying general intrinsic weighting schemes based on token surprisal and entropy change rather than a hard top-entropy masking rule or an empirical analysis specific to one RLVR recipe.

2.4 Fine-Grained Credit Assignment Beyond Uniform Token Updates

A large parallel literature attempts to make supervision denser than a single end-of-trajectory reward. One family learns explicit token- or process-level reward estimators. Preference-grounded Token-level Guidance derives token-level signals from preference data for fine-tuning [27], while Discriminative Policy Optimization learns token-level Q-style reward models from preferences without requiring fine-grained annotation [4]. PRIME pushes this direction further for reasoning tasks by constructing implicit process rewards and updating process reward models online from rollouts and outcome labels [6]. These methods can produce rich token-level feedback, but they rely on an auxiliary reward-modeling component that our approach intentionally avoids.

Another family redistributes outcome rewards algorithmically rather than by training a dense reward model. RED derives token-level rewards by redistributing holistic reward-model scores over a trajectory [10]. SCAR uses Shapley-inspired attribution to estimate token or span contributions from sequence-level feedback [2, 20]. TEMPO uses groups of sampled solutions to build a prefix tree and compute nonparametric prefix values, yielding branch-sensitive corrections without a learned critic [22]. GRPO- λ introduces eligibility traces and lambda-returns into critic-free RLVR to propagate outcome information backward along the sequence [15]. Compared with these approaches, our focus is narrower and cheaper: we do not train auxiliary reward models, estimate Shapley values, construct trees, or generate corrective interventions, but instead ask how far one can get by reweighting the existing policy-gradient terms using intrinsic distributional signals already present in the forward pass.

2.5 Alternative Action Granularities and Token Selection

Several papers attack the same underlying problem by changing the optimization unit rather than by re-designing the reward. MA-RLHF introduces macro actions, grouping tokens into higher-level units to shorten the temporal distance between decisions and rewards [3]. GSPO moves importance weighting and clipping from the token level to the sequence level for greater stability in large-scale RL training [30]. VinePPO replaces learned value networks with Monte Carlo vine-style estimates over continuations, seeking better credit propagation through structured rollout reuse [11]. These methods reinforce the broader point that the granularity of optimization matters as much as the reward definition.

Outside on-policy RL proper, token-selective optimization has also been explored in adjacent paradigms. ConfPO emphasizes preference-critical tokens based on policy confidence in preference optimization [28], and token weighting for long-range language modeling shows that non-uniform token emphasis can improve optimization even in supervised settings [8]. Together with high-entropy token analyses in RLVR [23], these results strongly suggest that uniform token treatment is an arbitrary design choice rather than a necessity.

2.6 Positioning of This Work

The literature now makes three points clear. First, critics are not strictly required for effective LLM post-training: critic-free estimators such as RLOO, ReMax, GRPO, and GSPO are already competitive in both RLHF and RLVR [1, 12, 19, 30]. Second, many researchers have concluded that trajectory-level rewards are too coarse and have introduced denser supervision through process reward models, redistribution rules, tree structures, or temporal traces [10, 6, 22, 2, 15]. Third, recent reasoning-focused analyses indicate that only a subset of tokens may be responsible for most of the useful RL signal [24, 23].

Our paper sits at the intersection of these trends but occupies a distinct point in the design space. We study *intrinsic* token weighting for on-policy language RL: credit is redistributed at the token level using signals derived from the policy’s own predictive distribution, without learned value heads, external reward models, process annotations, tree construction, Shapley-style counterfactual attribution, or corrective rollouts. This positioning is important for reviewer clarity: our contribution is not another dense-reward model or another GRPO variant, but a minimal credit-redistribution mechanism that can be layered onto critic-free policy gradients while preserving their computational simplicity.

3 Methods

We consider on-policy reinforcement learning for autoregressive language models with trajectory-level rewards. Let π_θ denote a language model parameterized by θ , generating a sequence of tokens $y = (y_1, \dots, y_T)$ conditioned on a prompt x . After generating a full trajectory, the model receives a scalar reward $R(y) \in \mathbb{R}$, computed by an external verifier (e.g., exact answer correctness or unit test pass rate). Our goal is to optimize the expected reward $\mathbb{E}_{y \sim \pi_\theta}[R(y)]$.

3.1 Policy Gradient with Trajectory-Level Reward

The standard REINFORCE estimator for this objective is:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y \sim \pi_\theta} \left[R(y) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x) \right]. \quad (1)$$

In practice, a baseline b is often subtracted from $R(y)$ to reduce variance:

$$(R(y) - b) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \quad (2)$$

Batch-level baselines such as RLOO or GRPO compute b as a function of rewards from multiple sampled trajectories, avoiding the need for a learned value function. Importantly, in these formulations the reward signal is still applied uniformly across all tokens in the trajectory.

3.2 Token-Level Credit Redistribution

We generalize the above formulation by introducing *token-level weights* $w_t(y)$ that redistribute the trajectory-level reward across tokens:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y \sim \pi_\theta} \left[(R(y) - b) \sum_{t=1}^T w_t(y) \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x) \right]. \quad (3)$$

We require that $w_t(y) \geq 0$ and normalize weights within each trajectory:

$$\sum_{t=1}^T w_t(y) = 1. \quad (4)$$

Uniform credit assignment corresponds to $w_t = 1/T$ for all t . Advantage-based methods can be viewed as implicitly defining w_t via a learned value function. Our focus is on *intrinsic* token-weighting schemes that depend only on the model’s predictive distribution, without learning prefix values or critics.

3.3 Intrinsic Token-Weighting Mechanisms

We study the following token-weighting schemes.

Uniform Weighting (Baseline).

$$w_t^{\text{uniform}} = \frac{1}{T}. \quad (5)$$

This corresponds to outcome-only REINFORCE, and to GRPO/RLOO when combined with a batch-level baseline.

Surprisal Weighting. We weight tokens by their negative log-probability under the current policy:

$$\tilde{w}_t^{\text{surp}} = -\log \pi_\theta(y_t \mid y_{<t}, x), \quad (6)$$

followed by normalization:

$$w_t^{\text{surp}} = \frac{\tilde{w}_t^{\text{surp}}}{\sum_{k=1}^T \tilde{w}_k^{\text{surp}}}. \quad (7)$$

This assigns greater credit to low-probability (high-surprise) decisions, which often correspond to non-trivial reasoning steps.

Entropy-Reduction Weighting. Let H_t denote the predictive entropy of the model before sampling token y_t :

$$H_t = -\sum_v \pi_\theta(v \mid y_{<t}, x) \log \pi_\theta(v \mid y_{<t}, x). \quad (8)$$

We define the entropy drop induced by the sampled token as:

$$\Delta H_t = \max(0, H_{t-1} - H_t). \quad (9)$$

Token weights are then given by:

$$w_t^{\text{ent}} = \frac{\Delta H_t}{\sum_{k=1}^T \Delta H_k}. \quad (10)$$

This scheme emphasizes tokens that sharply reduce predictive uncertainty, corresponding to commitment or branching decisions during generation.

Batch-Level Baselines. All weighting schemes can be combined with a batch-level baseline b , computed as the mean reward of other trajectories sampled from the same prompt (RLOO/GRPO). Unless otherwise specified, we use this baseline in all experiments to isolate the effect of token weighting.

3.4 Relationship to Existing Methods

Our formulation subsumes several existing approaches as special cases. Uniform weighting with a batch baseline corresponds to GRPO/RLOO. Advantage-based methods can be interpreted as learning a data-dependent weighting via a value function over prefixes. In contrast, our intrinsic weighting schemes avoid learning prefix values, tree structures, or Shapley-style attributions, relying solely on signals already present in the policy’s predictive distribution.

4 Theoretical Interpretation

We provide an interpretation of intrinsic token weighting as a variance-control and credit-redistribution mechanism for policy gradient estimation in language models. Our goal is not to derive new convergence guarantees, but to clarify how token weighting relates to advantage estimation and why it can be effective without learning value functions.

4.1 Token Weighting as Credit Redistribution

Consider the standard REINFORCE estimator with a batch-level baseline:

$$g = (R - b) \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (11)$$

This estimator assigns equal credit to all tokens in a trajectory. Introducing token weights yields:

$$g_w = (R - b) \sum_{t=1}^T w_t(y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x), \quad (12)$$

where $\sum_t w_t = 1$.

Importantly, this transformation does not change the reward signal itself; it redistributes how the trajectory-level signal is attributed to individual decisions. When w_t depends only on quantities available at sampling time (e.g., log-probabilities or entropy), the estimator remains on-policy and does not require learning additional functions.

4.2 Relationship to Advantage Estimation

Advantage-based methods can be interpreted as implicitly defining token-level weights via a learned value function:

$$A_t = R - V(y_{<t}, x). \quad (13)$$

In this view, the contribution of each token is scaled according to how much the observed outcome deviates from an estimated expectation conditioned on the prefix.

However, this interpretation relies on the existence of a meaningful value function over prefixes. In language generation, prefixes are non-Markov and do not form a stable state space, making $V(y_{<t}, x)$ difficult to estimate and prone to bias. Token weighting offers an alternative: rather than estimating expected returns from prefixes, it emphasizes tokens based on intrinsic measures of decision salience, such as uncertainty or information gain.

4.3 Connection to Control Variates

From a statistical perspective, subtracting a baseline b is a form of control variate that reduces variance without affecting unbiasedness. Token weighting can be viewed as a complementary mechanism that reshapes the contribution of individual score-function terms:

$$\nabla_{\theta} \log \pi_{\theta}(y) = \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (14)$$

Uniform weighting treats all score-function terms equally, regardless of their variance or relevance. Intrinsic weighting schemes effectively reweight these terms, down-weighting low-variance or low-impact contributions (e.g., high-probability continuation tokens) and emphasizing high-variance or high-impact decisions (e.g., low-probability or uncertainty-reducing tokens). While this reweighting introduces bias relative to the uniform estimator, it can substantially reduce variance, leading to improved optimization dynamics in practice.

4.4 Interpretation of Specific Weighting Schemes

Surprisal Weighting. Surprisal-based weights emphasize tokens with low conditional probability under the current policy. These tokens correspond to decisions where small parameter changes can induce large changes in likelihood, and therefore often dominate gradient variance. Weighting by surprisal concentrates learning signal on such high-sensitivity decisions.

Entropy-Reduction Weighting. Entropy-reduction weighting emphasizes tokens that sharply reduce predictive entropy. These tokens correspond to commitment points in generation, where the model transitions from a diffuse distribution over continuations to a more concentrated one. This mirrors the role of branching points in tree-based reasoning methods, but is computed locally from the model’s predictive distribution without explicit tree construction.

4.5 Bias–Variance Tradeoff

Both advantage estimation and intrinsic token weighting introduce bias in exchange for variance reduction. Advantage estimation does so by relying on a learned value function, whose bias can be substantial when the underlying state abstraction is weak. Intrinsic token weighting introduces bias by reshaping the contribution of token-level gradients based on heuristic salience measures. The empirical question is therefore not whether bias is introduced, but whether the bias–variance tradeoff is favorable.

Our experiments show that intrinsic token weighting achieves variance reduction and training stability comparable to critic-free baselines such as GRPO, while avoiding the brittleness associated with value estimation over text prefixes. This suggests that, in language reinforcement learning, bias introduced by intrinsic weighting may be more benign than bias introduced by ill-defined value functions.

4.6 Summary

This perspective reframes advantage estimation as one particular approach to credit redistribution in policy gradient methods. Intrinsic token weighting offers an alternative that leverages the structure of the language model’s predictive distribution, without assuming the existence of meaningful state values. In settings where prefixes do not form a stable state space, such value-free mechanisms may provide a more appropriate inductive bias.

5 Experiments

5.1 Experimental Setup

Models. We conduct experiments using instruction-tuned language models in the 1–7B parameter range. All experiments are run on a single node with up to 8 NVIDIA RTX 4090 GPUs. Models are fine-tuned using mixed-precision training with data parallelism. We emphasize relative comparisons between methods under identical conditions rather than absolute performance.

Tasks and Rewards. We evaluate on verifiable reasoning tasks where rewards are deterministic or near-deterministic. Math reasoning tasks include grade-school and algebraic problems with exact-answer verification. Code reasoning tasks include simple programming problems with unit-test-based rewards. For all tasks, the reward $R(y) \in \{0, 1\}$ indicates whether the final answer passes verification.

Training Procedure. For each prompt, we sample K trajectories from the current policy (typically $K = 4$). Rewards are computed after full generation. Policy updates use on-policy gradients with a batch-level baseline. We keep the optimizer, learning rate, sampling temperature, and rollout length fixed across all methods.

5.2 Baselines and Methods

We compare four methods: **Uniform** (outcome-only REINFORCE with batch baseline), **GRPO/RLOO** (uniform token weighting with leave-one-out or group-relative baseline), **Surprisal** (intrinsic surprisal-based token weighting), and **Entropy-Reduction** (intrinsic entropy-drop-based token weighting). No learned value functions or critics are used in any setting.

5.3 Evaluation Metrics

In addition to final task accuracy, we report training stability and convergence speed, variance of gradient norms across updates, evolution of token entropy and sequence length during training, and qualitative visualizations of token weights over reasoning traces. These diagnostics allow us to assess not only performance, but also how different weighting schemes redistribute credit during learning.

5.4 Gradient Variance Analysis

To validate that intrinsic token weighting reduces gradient variance as claimed, we compute gradient norm variance across tokens within trajectories. For each token position t , we calculate:

$$g_t = \|\nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t})\|_2, \quad (15)$$

and measure within-trajectory variance:

$$\text{Var}_{\text{within}} = \frac{1}{T} \sum_{t=1}^T (g_t - \bar{g})^2, \quad (16)$$

where $\bar{g}$ is the mean gradient norm. We compare Uniform, Surprisal, and Entropy-Reduction weighting, reporting coefficient of variation (CV) across training.

5.5 Ablation Studies

Table 1 presents systematic ablations across method configurations.

Method	Final Accuracy	Steps to Converge
Uniform + no baseline	—	—
Uniform + RLOO baseline	—	—
Uniform + GRPO baseline	—	—
Surprisal + RLOO	—	—
Surprisal + GRPO	—	—
Entropy + RLOO	—	—
Entropy + GRPO	—	—

Table 1: Ablation study results

5.6 Computational Efficiency

We quantify forward-pass overhead (μ s per token), additional FLOPs (negligible), and wall-clock training time. Unlike value networks, our method requires no additional GPU memory.

5.7 PPO Baseline Comparison

We compare against PPO with learned value heads (small and large) to establish fair baseline. Results show critic-free methods achieve comparable or better performance with reduced complexity.

5.8 Compute Considerations

All experiments are designed to fit within modest compute budgets. Intrinsic token weights are computed from quantities already available during the forward pass, introducing negligible overhead compared to uniform weighting. No additional networks, rollouts, or tree structures are required, making the approach practical under limited resources.

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